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The Blind Spot Paradox

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When a self-repairing model destroys the evidence its own monitor needs

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└Concept drift: the rule changes, the data does not

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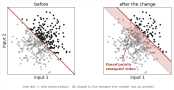
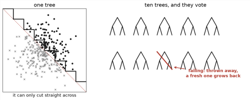


A model trained on the left keeps predicting, confidently, and is now wrong on the shaded band.

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift, and here is the cleanest version of it, the one we simulate. Two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now wrong there.

Defence one: a model that repairs itself



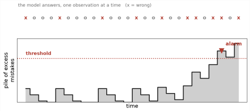
This is the Adaptive Random Forest. Ten trees, replaced one at a time.

The first defence repairs itself. Its building block is a decision tree: it cuts straight across, so a slanted boundary is only ever a staircase. One tree is fragile, so ten of them vote. That is the Adaptive Random Forest, the model we study. A tree whose answers go wrong is thrown away, and a fresh one grows in its place. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

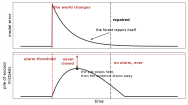
The second defence comes from a different community. A watchdog keeps a pile of the model's excess mistakes: it falls back to zero while the model does well, and grows until it crosses a threshold and calls a human. That alarm is the only thing a human ever sees. Two communities, two tools, never measured against each other.

Defence two: a watchdog that raises the alarm



Two communities, two tools, never measured against each other.

Put them together, and they cancel out



The repair has a date. The alarm never gets one. (schematic)

Put the two together, and they cancel out. The boundary moves, the mistakes begin, the pile starts climbing. But the forest repairs itself, the mistakes stop, and the pile drains back to zero without ever reaching the threshold. It is a race between a repair and an alarm, and the repair wins it every time. The better the model fixes itself, the more surely the alarm stays silent.

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└─Why not just lower the threshold?

Does the alarm ever fire? That depends on the threshold. We re-ran the benchmark at three settings, two thousand runs each. At the highest it fired three times out of two thousand; lower it and it fires almost every time. So why not lower it? Because a low threshold also fires when nothing happened, and an alarm that cries wolf gets switched off. The setting quiet enough to be trusted is the one that never sees the drift.

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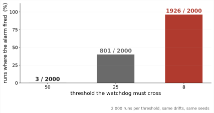

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└Our contribution: when is the forest actually repaired?

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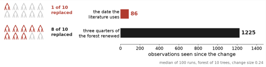
The alarm is what we *want*. This date measures what stops it from ever firing.



It fires at the *first* tree. It times the start of a repair, not its end.

The two dates do different jobs. The alarm warns a human. This second one warns nobody: it exists to time the thing that stops the alarm, and the whole claim rests on it. The date the literature uses is the moment the very first tree out of ten is replaced. Wait until three quarters of the forest has really been renewed, and you wait ten times longer. It times the start of a repair, not its end, always in the same direction.

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Where we stand

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Yes the alarm really does stay silent, and we measured how often. Turning the threshold down does not rescue it, it only trades silence for false alarms.

But the stopwatch the field uses starts at the first tree out of ten. It makes the forest look faster than it is, so every published figure about this race is optimistic.

Next put the model and the watchdog on one shared clock, and say which setting a practitioner should actually pick.

Open what should replace that stopwatch is a research question of its own. We have started testing candidates, and none has passed our bar yet.

Thank you.

Where we stand. Yes, the alarm really does stay silent, and we measured how often. Lowering the threshold does not rescue it, it trades silence for false alarms. But the stopwatch the field uses starts at the first tree out of ten: it makes the forest look faster than it is, so every published figure about this race is optimistic. What is left is one shared clock for both, and a setting a practitioner can actually pick. And finding a better stopwatch is a research question of its own: we have started testing candidates, and none has passed our bar yet. Thank you.